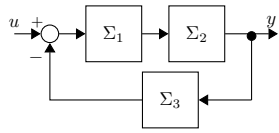


Modeling



$$y = \frac{\Sigma_2 \Sigma_1}{1 + \Sigma_2 \Sigma_1 \Sigma_3} u$$

Figure 1: Negative feedback loop.

State Space Representation

Non-Linear:

$$\dot{x}(t) = f(x(t), u(t))$$

$$y(t) = h(x(t), u(t))$$

LTI Systems:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

Linearization procedure:

1. Find (x_e, u_e) such that $f(x_e, u_e) = \dot{x}(t) = 0$
2. Calculate A, B, C, D matrices

$$A = \left. \frac{\partial f(x, u)}{\partial x} \right|_{(x_e, u_e)} \in \mathbb{R}^{n \times n} \quad B = \left. \frac{\partial f(x, u)}{\partial u} \right|_{(x_e, u_e)} \in \mathbb{R}^{n \times m}$$

$$C = \left. \frac{\partial h(x, u)}{\partial x} \right|_{(x_e, u_e)} \in \mathbb{R}^{p \times n} \quad D = \left. \frac{\partial h(x, u)}{\partial u} \right|_{(x_e, u_e)} \in \mathbb{R}^{p \times m}$$

System Classification:

- **Linear:** $f(x_1 + x_2) = f(x_1) + f(x_2)$, $f(a \cdot x) = a \cdot f(x)$, $f(0) = 0$
- **Static:** output depends only on present input
- **Causal:** output depends only on present and past inputs
- **Time-Invariant:** System input-output map does not change with time

Time Response

Initial condition response ($u(t) = 0$, $x(0) \neq 0$)
 $x_{IC}(t) = e^{At}x(0)$, $y_{IC}(t) = Ce^{At}x(0)$

Forced response ($u(t) \neq 0$, $x(0) = 0$)

$$x_F(t) = \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau,$$

$$y_F(t) = \int_0^t Ce^{A(t-\tau)} Bu(\tau) d\tau + Du(t)$$

Total response: $y(t) = y_{IC}(t) + y_F(t)$

Transfer Functions

Transfer Function Definition

$$y_{ss}(t) = G(s)e^{st}, \quad s \in \mathbb{C}$$

$$G(s) = C(sI - A)^{-1}B + D$$

$$(sI - A)^{-1} = \frac{\text{adj}(sI - A)}{\det(sI - A)}$$

Transfer Function Forms

POLYNOMIAL: $G(s) = \frac{b_m s^m + \dots + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_0}$

ROOT-LOCUS: $G(s) = \frac{k_{rl} \prod_{i=1}^m (s - z_i)}{s^q \prod_{i=1}^{n-q} (s - p_i)}$

BODE: $G(s) = \frac{k_{bode} \prod_{i=1}^m (\frac{s}{-z_i} + 1)}{s^q \prod_{i=1}^{n-q} (\frac{s}{-p_i} + 1)}$

PARTIAL FRACTION: $G(s) = \sum_{i=1}^n \frac{r_i}{s - p_i} + \sum_{j=1}^m \frac{r_j}{(s - p_j)^m}$

Residue Calculation

NON-REPEATED POLE: $r_i = \lim_{s \rightarrow p_i} (s - p_i)G(s)$

m-REPEATED POLE: $r_i = \frac{1}{(m-1)!} \lim_{s \rightarrow p_i} \frac{d^{m-1}}{ds^{m-1}} [(s - p_i)^m G(s)]$

Transfer Function $\iff$ State Space

PARTIAL FRACTION EXPANSION $A = \text{diag}(\lambda_1, \dots, \lambda_n)$,

$$G(s) = \sum_{i=1}^n \frac{p_i}{s - \lambda_i} + d \iff \begin{aligned} B &= [\sqrt{p_1} \quad \dots \quad \sqrt{p_n}]^\top, \\ C &= [\sqrt{p_1} \quad \dots \quad \sqrt{p_n}], \\ D &= d \end{aligned}$$

CONTROLLABLE CANONICAL FORM

$$G(s) = \frac{b_{n-1}s^{n-1} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0} + d$$

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix},$$

$$B = [0 \quad \dots \quad 0 \quad 1]^\top, \quad C = [b_0 \quad b_1 \quad \dots \quad b_{n-1}], \quad D = d$$

Canonical Inputs (Laplace Transform)

Input	$u(t)$	$U(s)$
Impulse	$\delta(t)$	1
Step	$u(t)$	$\frac{1}{s}$
Ramp (order q)	$t^q u(t)$	$\frac{q!}{s^{q+1}}$

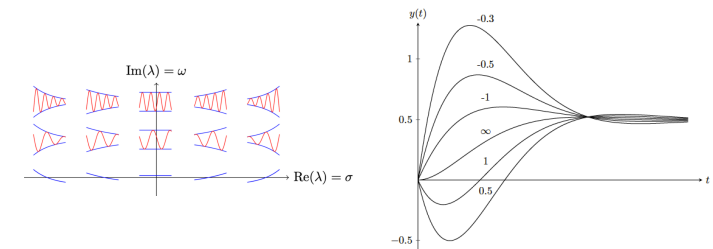
Initial Value Theorem: $\lim_{t \rightarrow 0} y(t) = \lim_{s \rightarrow \infty} sY(s)$

Final Value Theorem: $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$

Magnitude & Phase

Rule	Expression
Magnitude	$\left \frac{AB}{C} \right = \frac{ A B }{ C }, \quad A, B, C \in \mathbb{C}$
Phase	$\angle \left(\frac{AB}{C} \right) = \angle A + \angle B - \angle C$
Powers	$\angle \left(\frac{(a+jb)^x}{(c+jd)^y} \right) = x \angle(a+jb) - y \angle(c+jd)$
Complex angle	$\angle(a+jb) = \begin{cases} \arctan(b/a), & a > 0 \\ \arctan(b/a) + \pi, & a < 0, b \geq 0 \\ \arctan(b/a) - \pi, & a < 0, b < 0 \\ \frac{\pi}{2}, & a = 0, b > 0 \\ -\frac{\pi}{2}, & a = 0, b < 0 \end{cases}$

Effect of Poles and Zeros



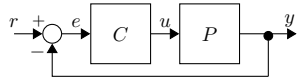
Root-Locus Sketching Rules

Rule	Description
Symmetry	Closed-loop poles are symmetric about the real axis
# of branches	# closed-loop poles = # open-loop poles
Start ($k \rightarrow 0$)	Closed-loop poles start at open-loop poles
End ($k \rightarrow \infty$)	If $\deg N = \deg D$: poles $\rightarrow$ zeros If $\deg N < \deg D$: excess poles $\rightarrow \infty$
Real-axis locus	For $k > 0$, locus exists to the left of an odd number of poles/zeros
Asymptotes	$\theta_q = \frac{(2q+1)180^\circ}{n-m}$, $q = 0, \dots, n-m-1$ $s_{\text{com}} = \frac{\sum_{i=1}^n \text{Re}(p_i) - \sum \text{Re}(z_i)}{n-m}$

THE MAGNITUDE RULE

$$|L(s)| = \frac{1}{|k|}, \quad \text{where } |L(s)| = |k_{rl}| \cdot \frac{\prod_{i=1}^m |s - z_i|}{\prod_{i=1}^n |s - p_i|}$$

Standard Feedback Architecture



$$\text{SENSITIVITY } (e \rightarrow y) \quad S(s) = \frac{1}{1+P(s)C(s)}$$

$$\text{COMPLEMENTARY SENSITIVITY } (r \rightarrow y) \quad T(s) = \frac{P(s)C(s)}{1+P(s)C(s)}$$

Dynamic Compensators

PI CONTROLLER	$C(s) = k_P + \frac{k_I}{s}$
PD CONTROLLER	$C(s) = k_D s + k_P$
PID CONTROLLER	$C(s) = k_D s + k_P + \frac{k_I}{s}$
LEAD COMPENSATOR	$C(s) = \frac{b}{a} \frac{s+a}{s+b}$, where $0 < a < b$
LAG COMPENSATOR	$C(s) = \frac{b}{a} \frac{s+a}{s+b}$, where $0 < b < a$

Steady-State Error

To achieve zero steady-state error to a ramp input of order q , the loop path from r to e must contain at least $q+1$ integrators.

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s S(s) R(s)$$

Time-Domain Specifications

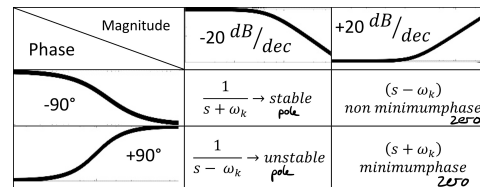
Standard Forms

1st order	$G(s) = \frac{1}{\tau s + 1}$
2nd order	$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ $= \frac{1}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta\frac{s}{\omega_n} + 1}$

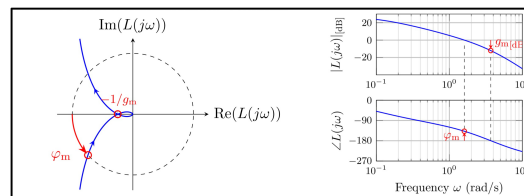
Time specifications:

Undamped 2nd order system	
Settling time to be within $d\%$ of y_{ss}	$T_d = -\frac{1}{ \sigma } \ln\left(\frac{d}{100}\right)$
Time to peak	$T_p = \frac{\pi}{\omega}$
Peak overshoot	$M_p = e^{\sigma\pi/\omega} = e^{\zeta\pi/\sqrt{1-\zeta^2}}$
Damping ratio from overshoot	$\zeta^2 = \frac{(\ln M_p)^2}{\pi^2 + (\ln M_p)^2}$
Rise time (0-100%)	$T_{100\%} = \frac{\frac{\pi}{2} - \phi}{\omega}$

Bode Plot



Factor		Magnitude	Phase
differentiator	$G(s) = s$	+20dB/dec	+90°
integrator	$G(s) = 1/s$	-20dB/dec	-90°
time delay	$G(s) = e^{-sT}$	0	-ωT



Nyquist

NYQUIST STABILITY CRITERION

$$Z = N + P$$

where Z is the number of unstable closed-loop poles, N is the net number of clockwise encirclements around $-1/k$ of the Nyquist plot, and P is the number of unstable open-loop poles.

Frequency Domain Specifications

Disturbance Rejection / Command Tracking: $S(j\omega)$ should be small at low frequencies.

Noise Rejection: $T(j\omega)$ should be small at high frequencies. Typically,

$$|S(j\omega)| \cdot |W_1(j\omega)| < 1$$

$$|T(j\omega)| \cdot |W_2(j\omega)| < 1$$

where $W_1(j\omega)$ and $W_2(j\omega)$ depend on the requirement.

Time delays

$$\text{APPROXIMATION: } e^{-sT} \approx \frac{2/T-s}{2/T+s}$$

$$\text{PHASE: } \angle G(j\omega) = -\omega T$$

$$\text{PHASE MARGIN: } \phi_m = \phi_{m,o} - \omega_c T$$

where $\phi_{m,o}$ is the phase margin of the open-loop system without time delay.

Limit Cycles

Limit cycles occur when $-\frac{1}{N(A)} = L(j\omega)$. A limit cycle is stable if an increase in amplitude causes results in entering a stable region of the Nyquist plot, and a decrease in amplitude results in entering an unstable region of the Nyquist plot.